



Figure 1: Purly the Snowman

We are in the last two weeks of our winter term. At the time of starting this last assignment, it is assumed that you have already completed all the previous assignments. At present, you should have a landing page for the game as shown in Figure 3. Also, your game character is now animation-enabled, which means it shows different animations with movement, rotation (not required if you are developing a 2D game) and jump movement. In addition, you should enable the 2D platform in the game as shown in Figure 4, where 'Purly' can move from one platform to another. Besides, two snow-guns will randomly try to throw snowballs in the opposite direction of the game arena. Every time, only one snow-gun will be active. Once the snowball reaches the opposite direction, it will be destroyed. And if it hits 'Purly', she will die.

The goal of Purly is to collect the yellow balloons, which are going to be placed on the platform. Balloons are going to be automatically placed at different positions on the platform. This means, the yellow balloons will be generated and placed on the platform with the help of background script. At the same time, for assignment 5, a black balloon will be generated and placed randomly on the platform. The ratio between yellow balloons and black balloons will be 5:1, which means that for every six balloons on the platform, five balloons will be yellow and one will be black. For collecting every yellow balloon, Purly will score 1 point, but for collecting one black balloon, Purly will get -3 points, which means 3 points will be deducted from the current score. So, Purly should try to avoid black balloons on her way.

Next, at present, we are at the beginning of Spring. Because of the weather change, we very often experience rainfall. Like that, in our game-playing environment, we are going to see rainfall from sky. The rainfall speed will be medium. At the same time, Purly is going to walk on a wet platform

surface. This will make water splash when Purly puts her step on the platform surface. This will make the game more real.

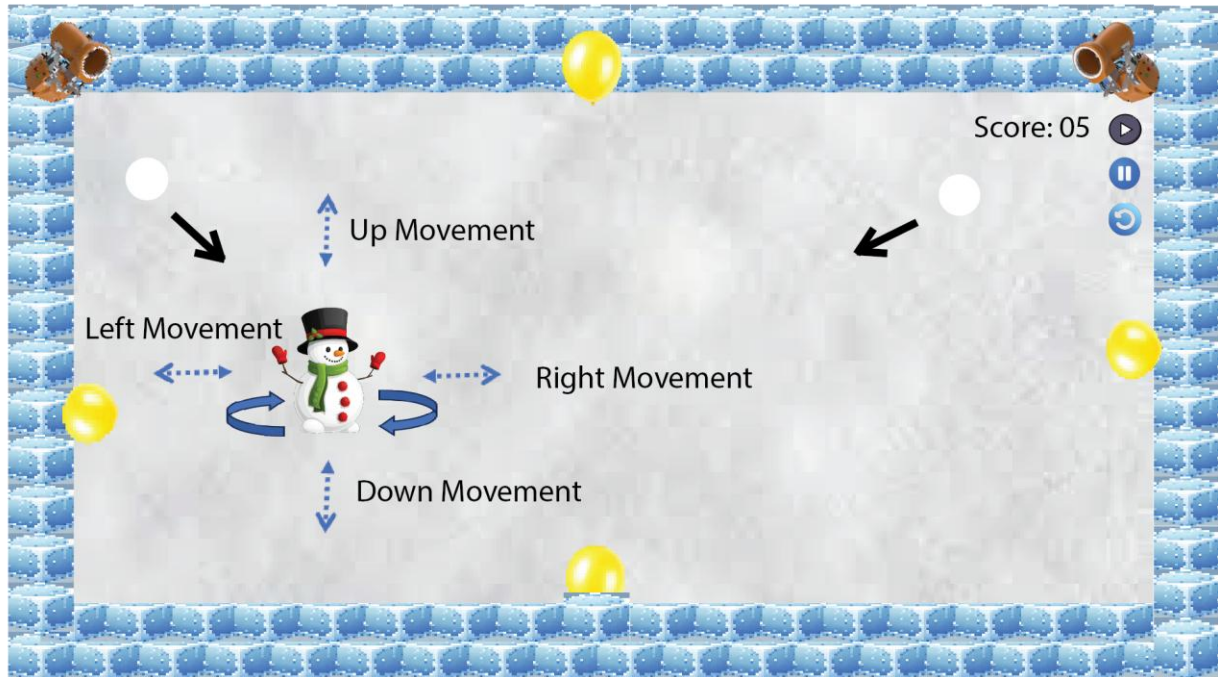


Figure 2: Game Arena

Finally, we know that the presence of sound effects for various movements and background music enhances the game experience. For our game, in this assignment, we are going to add some sound effects and background music. At the landing page, we will use music_1 and at the game-play environment, we will use music_2. And this music_2 will play in a loop until the game end. At the time of Purly jump on the wet platform, the splash sound effect will be activated. Also when Purly start jumping, the jump sound effect will be played. When Purly will collect the Yellow balloons, "Yellow Balloon" sound effect will be played. On the other hand, when Purly will collect the Black balloons by accident, "Black Balloon" sound effect will be played. When the current gameplay is over (Purly dies), "Game End" sound effect will be played. All these musics and sound effects together make the game-playing experience a top-notch one.

Please refer to the Figure 4 for the update game environment.



Figure 3: Loading Page

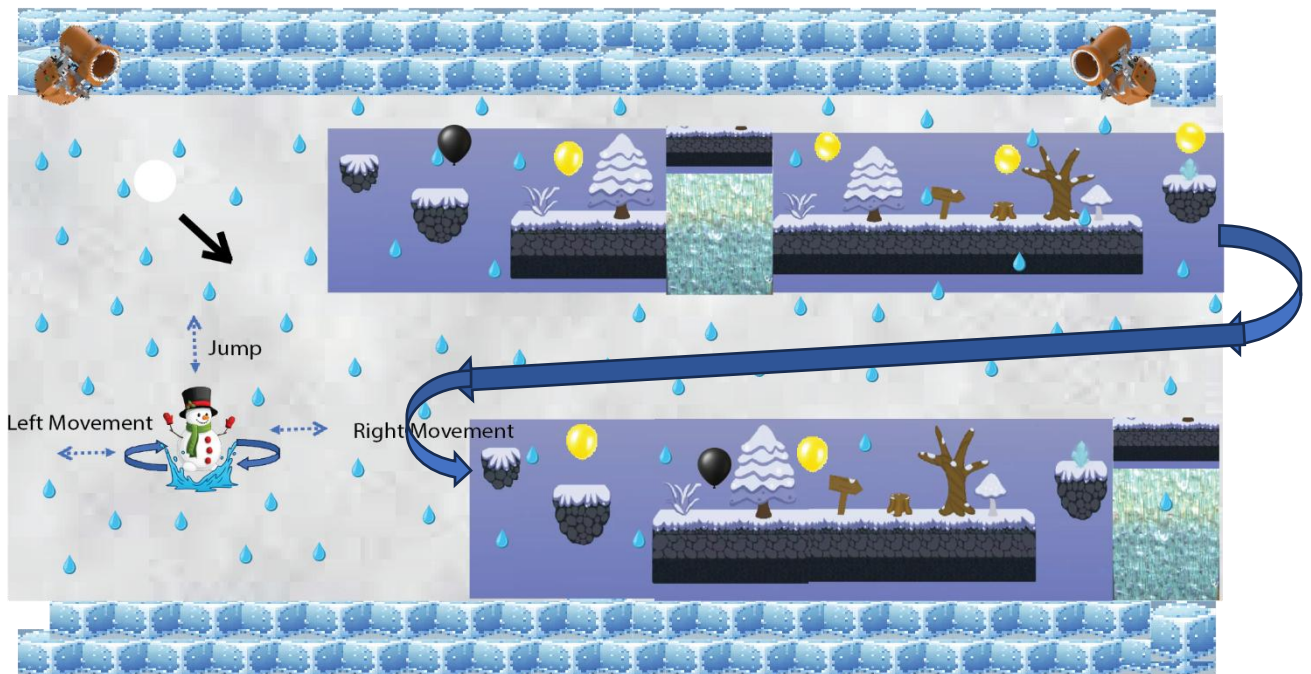


Figure 4: Updated Game Scene with Platform and Animated TilesMap

In this assignment you are going to do the following steps and will be evaluated based on the related evaluation marks:

Assignment Steps	Marks
a. Rigid-body based movement of the character "Purly". Purly will walk left, right, jump. To make the character's movement real, you need to enable proper animation for the character. Score calculation and visualization should be implemented and remain properly active in the game scenario. Also the game will start from the landing page and pause menu should be activated in the game playing environment.	10
b. Random generation of Black Balloon in the game scenario. Also, reduce 3 points when collecting the black balloon. Make sure to have one Black Balloon after every 5 yellow balloons.	10
c. Addition of music_1 in the landing page. The music will be played in a loop until the game starts. Also, whenever you toggle back to the landing page, music_1 will be played.	10
d. Addition of music_2 in the Game play environment and play in a loop until the game ends.	10
e. Addition of Jump sound effect with the character and be activated when "Purly" jumps.	5
f. Addition of water-splash_effect with the character and be activated when "Purly" lands on the platform surface.	5
g. Addition of Yellow_Collect_effect in the game and make it active when "Purly" will collect Yellow Balloons.	5
h. Addition of Black_Collect_effect in the game and make it active when "Purly" will collect Black Balloons.	5
i. Addition of Character_Died_Game_Over_effect in the game and make it active when "Purly" dies, either by getting hit with the ball or dropping from the platform.	5
j. Add the rainfall particle system in the gameplay window with the given raindrop. Make sure the rainfall frequency is medium.	15
k. Addition of the water splash particle system with the given water_splash image in the gameplay. It will be activated only when "Purly" will land on the platform. You need to control this splash particle system using a script and need to call it when you collide (land on) the surface (platform) after jumping. You can also take a small portion from the given water-splash image and use it with a pre-configured particle system effect to make it more realistic.	10
l. Lab work on Parallax in Unity	5
m. Lab work on Audio effect in Unity	5
Total	100

Submission Rules:

1. Omnivox/LEA might not allow you to upload all the game files. Your Google Drive/College provided one drive to store your summative assignment 05 and share the link with me. You can also upload your summative assignment in Github as well.
2. Before final submission, make a screen recording with all the implemented active features as instructed in this assignment.
3. Check off the assignment with your teacher in-person in class. Otherwise, you will get 0 for the submission.
4. Please respect the deadline. Students are asked to check off the work with the teacher the next week after the deadline is passed. Failure to do this will result in 0 even though you submitted the work properly.
5. Due date for this summative assessment is **17.05.25**, 17th May 2025. You can check-off the work with the teacher on **19th May, 2025**. But you need to record your game screen and **submit the recorded video in the Omnivox LEA under** summative assessment 05 by 17th May 2025. If you fail to do this, you will **automatically get 0**, even though you show your work to the teacher on 19th May.